Rovibrational analysis: Li_Omega

Sinc-DVR (Colbert–Miller) + sixth-degree Rydberg potential

Reduced mass $\mu = 12522.30592\ m_e$ (6.86949 amu). Grid: 500 sinc-DVR points on the CSV range. Level count is automatic: all bound states below D_e.

Table 1: Optimized sixth-degree Rydberg parameters (atomic units unless noted).

Parameter	Value
R_e (bohr)	3.314473
R_e (Å)	1.753944
D_e (hartree)	0.08139736
D_e (cm^{-1})	17864.656
D_e (kcal mol^{-1})	51.0776
$V(R_e)$ (hartree)	-0.08385014
c_1	0.64445265
c_2	-0.04457551
c_3	-0.00346911
c_4	0.00837446
c_5	-0.00100194
c_6	0.00006553
fit RMS (hartree)	8.493e-06

Table 2: Vibrational energy levels, spectrum $E_v - E_0$ and level differences ΔE_v (cm^{-1}).

v	E_v (J=0)	E_v (J=1)	$E_v - E_0$ (J=0)	ΔE_v (J=0)
0	198.0026	199.5931	0.0000	392.7591
1	590.7617	592.3422	392.7591	388.0596
2	978.8212	980.3916	780.8186	383.3352
3	1362.1564	1363.7164	1164.1538	378.5857
4	1740.7420	1742.2914	1542.7395	373.8109
5	2114.5530	2116.0914	1916.5504	369.0107
6	2483.5636	2485.0910	2285.5611	364.1849
7	2847.7485	2849.2645	2649.7460	359.3334
8	3207.0819	3208.5862	3009.0793	354.4561
9	3561.5380	3563.0304	3363.5354	349.5530
10	3911.0910	3912.5711	3713.0884	344.6240
11	4255.7149	4257.1826	4057.7124	339.6692
12	4595.3841	4596.8391	4397.3816	334.6887
13	4930.0728	4931.5148	4732.0703	329.6826
14	5259.7555	5261.1842	5061.7529	324.6512
15	5584.4067	5585.8218	5386.4041	319.5947
16	5904.0014	5905.4027	5705.9988	314.5135
17	6218.5150	6219.9021	6020.5124	309.4081
18	6527.9231	6529.2958	6329.9205	304.2790

v	E_v (J=0)	E_v (J=1)	$E_v - E_0$ (J=0)	ΔE_v (J=0)
19	6832.2021	6833.5601	6634.1995	299.1268
20	7131.3289	7132.6720	6933.3263	293.9525
21	7425.2814	7426.6092	7227.2788	288.7568
22	7714.0382	7715.3504	7516.0356	283.5408
23	7997.5790	7998.8753	7799.5764	278.3059
24	8275.8849	8277.1651	8077.8823	273.0533
25	8548.9382	8550.2020	8350.9356	267.7848
26	8816.7230	8817.9701	8618.7204	262.5021
27	9079.2251	9080.4552	8881.2225	257.2073
28	9336.4324	9337.6452	9138.4298	251.9026
29	9588.3349	9589.5303	9390.3324	246.5906
30	9834.9256	9836.1032	9636.9230	241.2742
31	10076.1998	10077.3594	9878.1972	235.9566
32	10312.1564	10313.2977	10114.1538	230.6410
33	10542.7974	10543.9203	10344.7948	225.3314
34	10768.1288	10769.2330	10570.1263	220.0319
35	10988.1607	10989.2461	10790.1582	214.7469
36	11202.9077	11203.9739	11004.9051	209.4812
37	11412.3889	11413.4360	11214.3863	204.2401
38	11616.6290	11617.6567	11418.6264	199.0288
39	11815.6578	11816.6660	11617.6552	193.8534
40	12009.5112	12010.4998	11811.5086	188.7197
41	12198.2309	12199.1999	12000.2283	183.6341
42	12381.8650	12382.8144	12183.8624	178.6031
43	12560.4681	12561.3978	12362.4656	173.6332
44	12734.1014	12735.0114	12536.0988	168.7311
45	12902.8324	12903.7229	12704.8299	163.9032
46	13066.7356	13067.6065	12868.7330	159.1560
47	13225.8916	13226.7431	13027.8890	154.4955
48	13380.3871	13381.2193	13182.3845	149.9276
49	13530.3147	13531.1278	13332.3121	145.4575
50	13675.7722	13676.5665	13477.7696	141.0900
51	13816.8622	13817.6379	13618.8596	136.8291
52	13953.6913	13954.4486	13755.6887	132.6781
53	14086.3694	14087.1086	13888.3669	128.6397
54	14215.0091	14215.7305	14017.0066	124.7156
55	14339.7247	14340.4287	14141.7222	120.9068
56	14460.6315	14461.3183	14262.6289	117.2134
57	14577.8450	14578.5150	14379.8424	113.6351
58	14691.4800	14692.1336	14493.4775	110.1704
59	14801.6505	14802.2880	14603.6479	106.8177
60	14908.4682	14909.0901	14710.4656	103.5747
61	15012.0429	15012.6495	14814.0404	100.4387
62	15112.4817	15113.0732	14914.4791	97.4067

v	E_v (J=0)	E_v (J=1)	$E_v - E_0$ (J=0)	ΔE_v (J=0)
63	15209.8883	15210.4653	15011.8858	94.4755
64	15304.3638	15304.9265	15106.3612	91.6417
65	15396.0055	15396.5544	15198.0030	88.9022
66	15484.9077	15485.4430	15286.9051	86.2535
67	15571.1612	15571.6833	15373.1586	83.6924
68	15654.8537	15655.3628	15456.8511	81.2160
69	15736.0696	15736.5662	15538.0671	78.8212
70	15814.8908	15815.3752	15616.8883	76.5054
71	15891.3962	15891.8686	15693.3936	74.2659
72	15965.6621	15966.1229	15767.6595	72.1005
73	16037.7626	16038.2120	15839.7600	70.0069
74	16107.7695	16108.2079	15909.7669	67.9831
75	16175.7525	16176.1802	15977.7500	66.0270
76	16241.7795	16242.1967	16043.7770	64.1368
77	16305.9164	16306.3234	16107.9138	62.3107
78	16368.2271	16368.6242	16170.2245	60.5468
79	16428.7739	16429.1614	16230.7713	58.8432
80	16487.6171	16487.9952	16289.6145	57.1979
81	16544.8150	16545.1840	16346.8124	55.6090
82	16600.4239	16600.7841	16402.4213	54.0742
83	16654.4981	16654.8498	16456.4956	52.5915
84	16707.0896	16707.4329	16509.0870	51.1583
85	16758.2479	16758.5831	16560.2453	49.7723
86	16808.0203	16808.3475	16610.0177	48.4309
87	16856.4512	16856.7707	16658.4486	47.1315
88	16903.5827	16903.8947	16705.5801	45.8713
89	16949.4540	16949.7587	16751.4514	44.6484
90	16994.1024	16994.4000	16796.0998	43.4662
91	17037.5686	17037.8593	16839.5660	42.3651
92	17079.9337	17080.2182	16881.9311	41.5290
93	17121.4627	17121.7434	16923.4601	41.4052
94	17162.8679	17163.1499	16964.8653	–

Table 3: Spectroscopic constants (cm^{-1}), finite differences of the first levels (Eq. 7 of Silva *et al.*, J. Mol. Model. **24**, 235, 2018).

Constant	J=0	J=1
ω_e	397.4348	397.4251
$\omega_e x_e$	2.33110	2.33122
$\omega_e y_e$	-4.1464e-03	-4.1470e-03
B_e	0.797689	
α_e	4.8425e-03	
γ_e	-5.9286e-05	

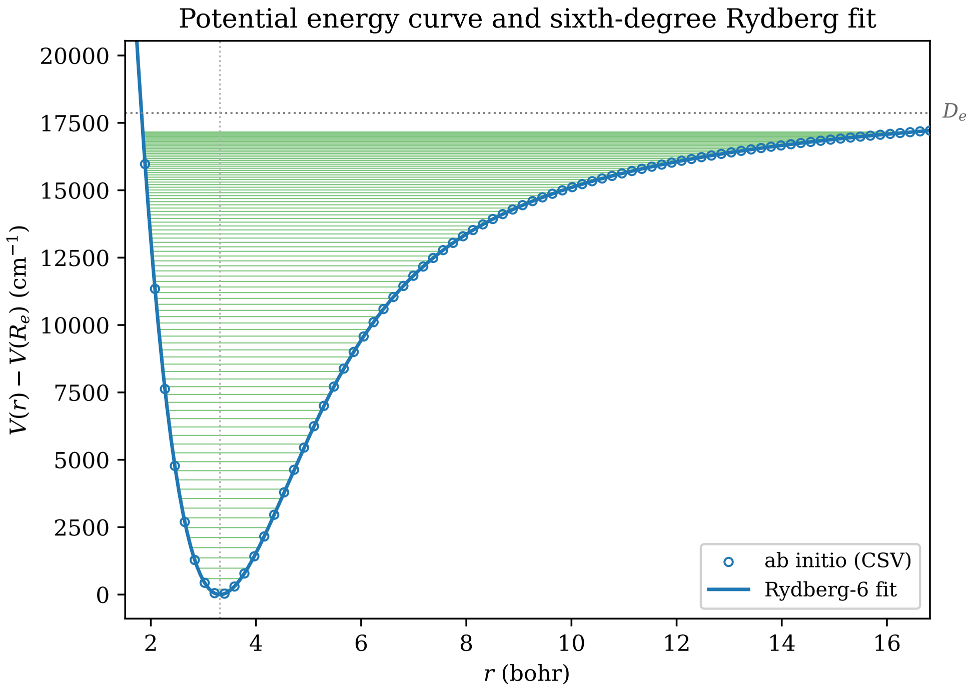

Figure 1: Potential energy curve: ab initio points, sixth-degree Rydberg fit, and the J=0 vibrational levels (green).

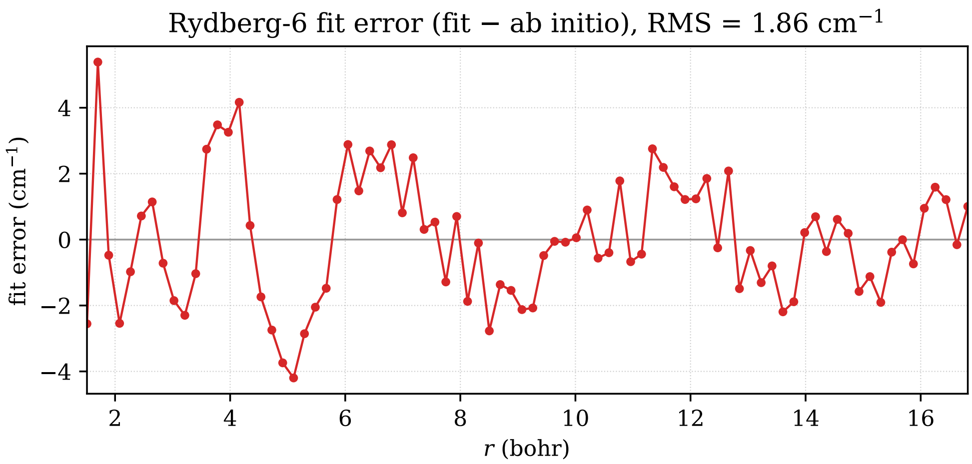

Figure 2: Residual of the sixth-degree Rydberg fit, fit − ab initio, over the whole grid.

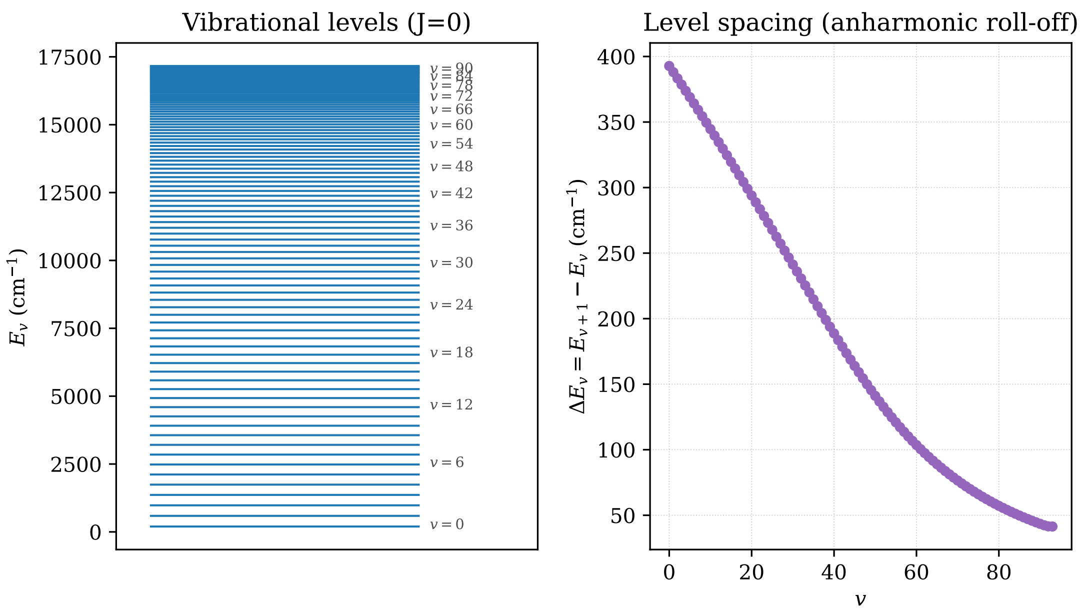

Figure 3: Vibrational spectrum: energy-level ladder (left) and the anharmonic decrease of the level spacing ΔE_v (right).